

MIT OpenCourseWare

14.04

Intermediate Microeconomic Theory 14.04 — Intermediate Microeconomic Theory May

2026 **EXTREMELY HARD – MIT LEVEL****Time Limit: 3 hours****Total Marks: 100**

1. (5 points) Construct an original proof-level problem involving Utility Maximization for 14.04 (Intermediate Microeconomic Theory) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
2. (4 points) Analyze the limitations of standard approaches to General Equilibrium when applied to edge cases in 14.04. Construct explicit counterexamples showing where intuitive reasoning fails.
3. (3 points) Construct an original proof-level problem involving Welfare for 14.04 (Intermediate Microeconomic Theory) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
4. (2 points) Analyze the limitations of standard approaches to Externalities when applied to edge cases in 14.04. Construct explicit counterexamples showing where intuitive reasoning fails.
5. (6 points) Construct an original proof-level problem involving Public Goods for 14.04 (Intermediate Microeconomic Theory) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
6. (4 points) Prove the fundamental theorem concerning Information Economics in the context of 14.04 (Intermediate Microeconomic Theory). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
7. (6 points) Construct an original proof-level problem involving Utility Maximization for 14.04 (Intermediate Microeconomic Theory) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
8. (4 points) Prove the fundamental theorem concerning General Equilibrium in the context of 14.04 (Intermediate Microeconomic Theory). State the theorem precisely, provide a complete proof, and discuss three important corollaries.

9. (5 points) Analyze the limitations of standard approaches to Welfare when applied to edge cases in 14.04. Construct explicit counterexamples showing where intuitive reasoning fails.
10. (6 points) Prove the fundamental theorem concerning Externalities in the context of 14.04 (Intermediate Microeconomic Theory). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
11. (4 points) Analyze the limitations of standard approaches to Public Goods when applied to edge cases in 14.04. Construct explicit counterexamples showing where intuitive reasoning fails.
12. (5 points) Prove the fundamental theorem concerning Information Economics in the context of 14.04 (Intermediate Microeconomic Theory). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
13. (5 points) Prove the fundamental theorem concerning Utility Maximization in the context of 14.04 (Intermediate Microeconomic Theory). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
14. (2 points) Analyze the limitations of standard approaches to General Equilibrium when applied to edge cases in 14.04. Construct explicit counterexamples showing where intuitive reasoning fails.
15. (4 points) Prove the fundamental theorem concerning Welfare in the context of 14.04 (Intermediate Microeconomic Theory). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
16. (3 points) Prove the fundamental theorem concerning Externalities in the context of 14.04 (Intermediate Microeconomic Theory). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
17. (5 points) Analyze the limitations of standard approaches to Public Goods when applied to edge cases in 14.04. Construct explicit counterexamples showing where intuitive reasoning fails.
18. (7 points) Prove the fundamental theorem concerning Information Economics in the context of 14.04 (Intermediate Microeconomic Theory). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
19. (5 points) Prove the fundamental theorem concerning Utility Maximization in the context of 14.04 (Intermediate Microeconomic Theory). State the theorem precisely, provide a complete proof, and discuss three important corollaries.

20. (2 points) Construct an original proof-level problem involving General Equilibrium for 14.04 (Intermediate Microeconomic Theory) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
21. (3 points) Prove the fundamental theorem concerning Welfare in the context of 14.04 (Intermediate Microeconomic Theory). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
22. (2 points) Prove the fundamental theorem concerning Externalities in the context of 14.04 (Intermediate Microeconomic Theory). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
23. (2 points) Analyze the limitations of standard approaches to Public Goods when applied to edge cases in 14.04. Construct explicit counterexamples showing where intuitive reasoning fails.
24. (3 points) Analyze the limitations of standard approaches to Information Economics when applied to edge cases in 14.04. Construct explicit counterexamples showing where intuitive reasoning fails.
25. (3 points) Prove the fundamental theorem concerning Utility Maximization in the context of 14.04 (Intermediate Microeconomic Theory). State the theorem precisely, provide a complete proof, and discuss three important corollaries.

Solutions

Solution 1:

Solution approach: First recall the definitions and key theorems related to Utility Maximization from 14.04. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 2:

The edge case analysis reveals the subtle assumptions underlying standard treatments of General Equilibrium. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 3:

Solution approach: First recall the definitions and key theorems related to Welfare from 14.04. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 4:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Externalities. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 5:

Solution approach: First recall the definitions and key theorems related to Public Goods from 14.04. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 6:

The proof follows from the definitions and properties established in 14.04. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 7:

Solution approach: First recall the definitions and key theorems related to Utility Maximization from 14.04. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 8:

The proof follows from the definitions and properties established in 14.04. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 9:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Welfare. By constructing explicit counterexamples, we see that the usual hypotheses

cannot be weakened without loss of the theorem's conclusion.

Solution 10:

The proof follows from the definitions and properties established in 14.04. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 11:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Public Goods. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 12:

The proof follows from the definitions and properties established in 14.04. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 13:

The proof follows from the definitions and properties established in 14.04. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 14:

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Solution 20:

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Solution 21:

The proof follows from the definitions and properties established in 14.04. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 22:

The proof follows from the definitions and properties established in 14.04. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 23:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Public Goods. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 24:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Information Economics. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 25:

The proof follows from the definitions and properties established in 14.04. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.